

1 Introduction

How large can a convex body’s symplectic capacity be if its volume is fixed? Viterbo’s conjecture proposed that a ball gives the largest value [Vit00]. For the Ekeland–Hofer–Zehnder capacity in dimension four, the proposed inequality was

$$\text{sys}(K) := \frac{c_{\text{EHZ}}(K)^2}{2 \text{vol}_4(K)} \leq 1.$$

The quantity on the left is called the systolic ratio. Capacity scales by the square of a dilation and four-dimensional volume by its fourth power, so this ratio compares shape independently of overall size. Translations and linear symplectic maps also preserve it.

The inequality is false. Haim–Kislev and Ostrover found a counterexample formed from two regular pentagons in complementary Lagrangian planes [HO26]. Writing the coordinates as $(q, p) \in \mathbb{R}^2 \oplus \mathbb{R}^2$, one representative is

$$K_{\text{HKO}} = P_5 \times R(-\pi/2)P_5, \quad \text{sys}(K_{\text{HKO}}) = \frac{3 + \sqrt{5}}{5} > 1.$$

Here P_5 is a centred regular pentagon with a vertex on the positive horizontal axis, and $R(\theta)$ is planar rotation. A product placed in the q - and p -planes is called a Lagrangian product. The relative position of its factors affects capacity even though it does not affect volume.

The counterexample leaves several geometric questions. Is the HKO body a local maximum of the systolic ratio among convex bodies? Are there other families with large ratios? Can the computations reveal geometric relationships involving capacity? This thesis investigates these questions through local analysis, finite algorithms and data science. This is the sense in which we continue to probe Viterbo’s conjecture: although its proposed bound is false, the geometric relationship between symplectic capacity and volume remains to be understood.

The mathematical results identify two ways of changing the counterexample that cannot improve its ratio: sufficiently small ten-facet perturbations and arbitrary independent linear deformations of its regular-pentagon factors. They also exhibit a distinct non-HKO local maximum in the same ten-facet space, at ratio one. These are different restrictions and examples: the first is local in the full ten-facet space, the second is global within a product family, and the third is a separate product body with a non-strict local maximum. Finite capacity formulas make these questions accessible to proof. The empirical investigation asks whether geometric measurements can explain capacity in broader populations and guide the search for further examples.

References

- [HO26] Pazit Haim-Kislev and Yaron Ostrover. “A counterexample to Viterbo’s conjecture”. In: *Annals of Mathematics* 203.2 (2026), pp. 603–622. DOI: 10.4007/annals.2026.203.2.5.
- [Vit00] Claude Viterbo. “Metric and isoperimetric problems in symplectic geometry”. In: *Journal of the American Mathematical Society* 13.2 (2000), pp. 411–431. DOI: 10.1090/S0894-0347-00-00328-3.